

Samba Gangineni

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EDUCATION

BOSTON UNIVERSITY

MSC IN COMPUTER INFO SYSTEMS

Grad. Jan 2020

GPA: **3.86 / 4.0**

COURSEWORK

Big Data Analytics with Spark

Artificial Intelligence

DataMining & Business Intelligence

R.V.R & J.C.C.E

B.TECH IN CIVIL ENGINEERING

Grad. May 2017

GPA: **9.23 / 10.0**

COURSEWORK

Programming with C + Practicum

Database Management System

Computer Programming in CEngg

(Teaching Asst)

SKILLS

PROGRAMMING

Javascript • Kotlin • Python • Java •
React • Flask • Go • SQL • bash •
HTML • CSS • HCL •

DATABASES

MongoDB • MSSQL Server •

FRAMEWORKS & TOOLS

Temporal • Spring Boot • Spark •
TensorFlow • Rabbitmq • Docker •
Kubernetes • Helm • Terraform •
Pulumi • AWS • Git • Jenkins •

LINKS

LinkedIn: [in/samba-gangineni](https://www.linkedin.com/in/samba-gangineni)

Github: [samba-gangineni](https://github.com/samba-gangineni)

DevPost: [samba693](https://devpost.com/users/samba693)

StackOverflow: [users/8721887/](https://stackoverflow.com/users/8721887/)

SPORTS

2016	College, Chess	1 st place
2016	University, Chess	1 st place
2015	College, Chess	3 rd place

VOLUNTEER

2016	RVRJC, Sports	Organizer
2016	RVRJC, Fest	Organizer

EXPERIENCE

AVAILITY | SOFTWARE ENGINEER IV

Feb 2020 – Present | Boston, MA

- Designed and implementing a microservices-based stateless product pipeline with event-driven architecture on Temporal, aiming to make data storage optional and eliminate MongoDB.
- Led the successful transition of infrastructure code and customer environments from Pulumi to Terraform, improving infrastructure-as-code management and cloud infrastructure stability.
- Achieved a 5x performance improvement in FHIR data deduplication, minimized database and RabbitMQ load, and ensured high data accuracy by introducing a redesigned architecture and Kotlin coroutines, effectively reducing lock contentions and system bottlenecks.
- Streamlined FHIR data ingestion using Temporal, managing CI/CD and deployment into EKS using Terraform to drive efficient product integration and scalable healthcare data processing.
- Spearheaded MongoDB sharding for a large-scale healthcare data, redesigning application queries to boost performance and regenerating 6 billion FHIR resources to support 10 million patients.
- Managed large-scale claims data ingestion leveraging Spark, AWS Lambda, EMR, and Kubernetes, delivering efficient and fault-tolerant data processing for a population of 13 million patients.
- Built a clinical inference model to detect gaps in patient data, improving data integrity and supporting better health outcomes.
- Created realistic clinical synthetic notes using GPT-2 models and classifiers, simulating actual patient notes to support rigorous product testing, refinement.
- Automated terminology updates, enabling data normalization and actionable insights for informed decision-making, particularly during COVID-19.

DIAMETER HEALTH | INFORMATICS AND ENGINEERING INTERN

Jun 2019 – Aug 2019 | Boston, MA

- Developed a transpiler in golang that converts clinical quality language into JavaScript, automating boilerplate code generation and streamlining development.
- Designed and implemented a synthetic data clinical HL7 message generator in JavaScript, providing a tool to improve product quality and accelerate testing and development cycles

RESEARCH

ACM DEBS 2019 | DISTRIBUTED AND EVENT BASED SYSTEMS

- Architected a real-time object detection system for LiDAR data using clustering and convolutional neural networks.
- <https://doi.org/10.1145/3328905.3330297>

AWARDS

2019	ACM DEBS Grand challenge winner (Object recognition)
2018	1st place in BigRedHacks (ResQu - An SOS application)
2018	3rd place in Att Hacks (Imound - An app for visually challenged)